

Total No. of Questions : 8]

SEAT No. :

PE-2745

[Total No. of Pages : 2

[6583]-326

T.E. (Mechanical)

**FOUNDATIONS OF SYSTEMS AND SYSTEMS
ENGINEERING**

**(Honors of Systems and Engineering)
(2019 Pattern) (Semester - V) (302041 MJ)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right side indicate full marks.

Q1) a) Explain Systems Science and illustrate the relationship between the systems science, System thinking and General system approach as applied to Engineered system. [9]

b) List and explain the life cycle stages of the system with a sketch. [9]

OR

Q2) a) Write short notes on system complexity and Emergence in System Engineering. [9]

b) Illustrates the relationship of the System of Interest (SOI), enabling systems, and the other systems in the operational environment. [9]

Q3) a) Explain the architecture definition process with an IPO diagram. [9]

b) State and explain the types of System Architecture. [8]

OR

Q4) a) Write short notes on Architectural views. [8]

b) Write short notes on Architectur Development and Traceability. [9]

Q5) a) What is MBSE and explain the evolution of MBSE. [9]

b) Explain and demonstrate the purpose of System modelling language-SysML. [9]

OR

P.T.O.

- Q6)** a) With a neat sketch explain the basic structure of SysML. [9]
b) Write a short notes on Modelling, simulation and trade-off analysis. [9]

- Q7)** a) Demonstrate the concept of Simulation modelling in the SE lifecycle. [8]
b) Write short notes on the basis of quantitative modelling. [9]

OR

- Q8)** a) Explain the role of modelling in decision making. [8]
b) Write a short notes on System dynamics modelling. [9]

